Cyber Risk Assessment & Threat Intelligence Platform

Project Overview

The Cyber Risk Intelligence Platform is a security analysis system designed to scan a target domain or IP address and identify potential vulnerabilities.

The system combines:

- Network scanning (Nmap)
- Threat intelligence (Virus Total)
- Risk scoring algorithms

It provides a visual dashboard, email alerts, and PDF reports to help users understand the security posture of a system.

Objectives

- Detect open ports and running services
- Analyze exposure and threat levels
- Calculate risk score and severity
- Provide actionable recommendations
- Visualize results using an interactive dashboard
- Send alerts for critical vulnerabilities
- Generate downloadable reports

Features

🔍 **Scanning**

- Scan top 1000 ports using Nmap
- Identify services, protocols, versions

🌐 **Threat Intelligence**

- Fetch data from Virus Total
- Detect malicious/suspicious IP reputation

📊 **Risk Analysis**

- Exposure Score (based on ports)
- Threat Score (based on VT data)
- Context Score

- Final Risk Score

📊 **Dashboard**

- KPI metrics (Risk, Threat, Exposure)
- Pie chart (Risk level)
- Bar chart (Score breakdown)
- Heatmap (Risk intensity)
- Services list

📜 **History & Trends**

- Store previous scans in SQLite
- Track risk over time

📧 **Email Alerts**

- Send alert if risk is HIGH or CRITICAL

📄 **PDF Report**

- Generate structured security report

System Architecture

User Input (Target Domain/IP)

↓

Nmap Scan

↓

Parse Scan Results

↓

Virus Total API Data

↓

Data Normalization

↓

Risk Calculation

↓

Dashboard	Email Alerts	PDF Report

Project Structure

```
cyberisk_platform/
│
├── dashboard/
│   └── app.py
│
├── modules/
│   ├── scanner.py
│   ├── normalizer.py
│   ├── analyser.py
│   ├── database.py
│   ├── emailer.py
│   └── pdf_generator.py
├── scan_results/
├── requirements.txt
├── README.md
└── DOCUMENTATION.md
```

Modules Explanation

scanner.py

- Runs Nmap scan
- Parses XML output
- Extracts:
  - IP
  - Ports
  - Services
  - Versions
- Fetches VirusTotal data

normalizer.py

- Cleans Virus Total data
- Converts raw data into structured format

analyser.py

- Calculates:
  - Exposure Score
  - Threat Score
  - Context Score
  - Risk Score
- Assigns severity (Low, Medium, High, Critical)

database.py

- Stores scan results in SQLite
- Retrieves historical data

emailer.py

- Sends alerts using SMTP (Gmail)
- Triggered when risk is HIGH/CRITICAL

pdf_generator.py

- Generates PDF report using FPDF
- Includes:
  - Target info
  - Scores
  - Recommendations

TECH STACK

Component	Technology
Language	Python
UI	Streamlit
Visualization	Plotly
Scanning	Nmap
Threat Intel	VirusTotal API
Database	SQLite
Email	SMTP (Gmail)
Reports	FPDF

How to Run

Step 1: Install dependencies

```
pip install -r requirements.txt
```

Step 2: Install Nmap

Download from: https://nmap.org/download.html

Step 3: Set environment variables

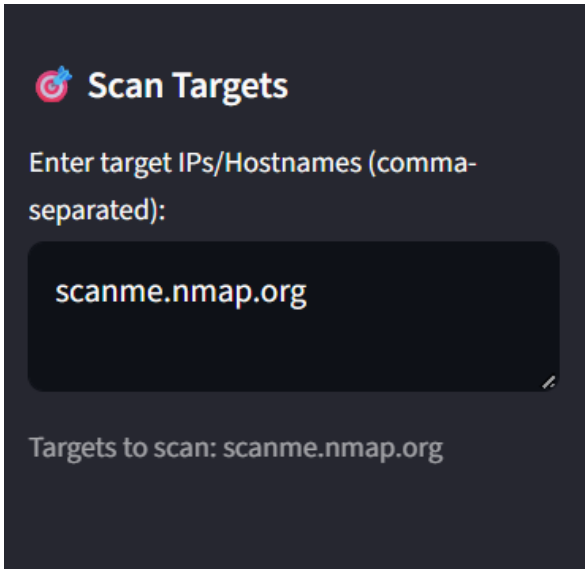
```
export VT_API_KEY="your_api_key"
export GMAIL_SENDER="your_email@gmail.com"
export GMAIL_PASSWORD="app_password"
export GMAIL_RECIPIENT="receiver_email"
```

Step 4: Run application

```
streamlit run dashboard/app.py
```

Sample Input

scanme.nmap.org



Sample Output

- Target IP detected
- Open ports listed
- Risk score calculated
- Severity displayed
- Charts: Pie chart, Bar graph, Heatmap
- Services detected , Email alert triggered (if critical or high) , PDF generated.

Security Analysis

What was found, how bad it is, and what to do about it

Security Posture: MODERATE — Review findings

Hosts Scanned	Open Ports	Critical Hosts	High Risk Hosts	VT Flagged
1	5	0	0	5

Key Findings

- Unencrypted (HTTP) services found without HTTPS alternative on 1 entries. Credentials or sensitive data may be transmitted in plaintext.
- 1 IP(s) flagged malicious by VirusTotal: 45.33.32.156. Investigate immediately.
- Remote access services exposed (ssh on 45.33.32.156) — secure with VPN, strong MFA, and restrict access where possible.

Immediate Actions

Sorted by risk score — highest priority first. Click to expand for details.

[Medium] **45.33.32.156:22** (ssh) — Risk Score: **4.3**

What to do: Disable password login — use SSH keys only. Consider changing to a non-standard port and implement rate limiting for connection attempts.

[Medium] **45.33.32.156:80** (http) — Risk Score: **4.3**

What to do: Check for directory listing, default credentials, and outdated web server software. Ensure only necessary content is exposed.

[Medium] **45.33.32.156:53** (domain) — Risk Score: **3.9**

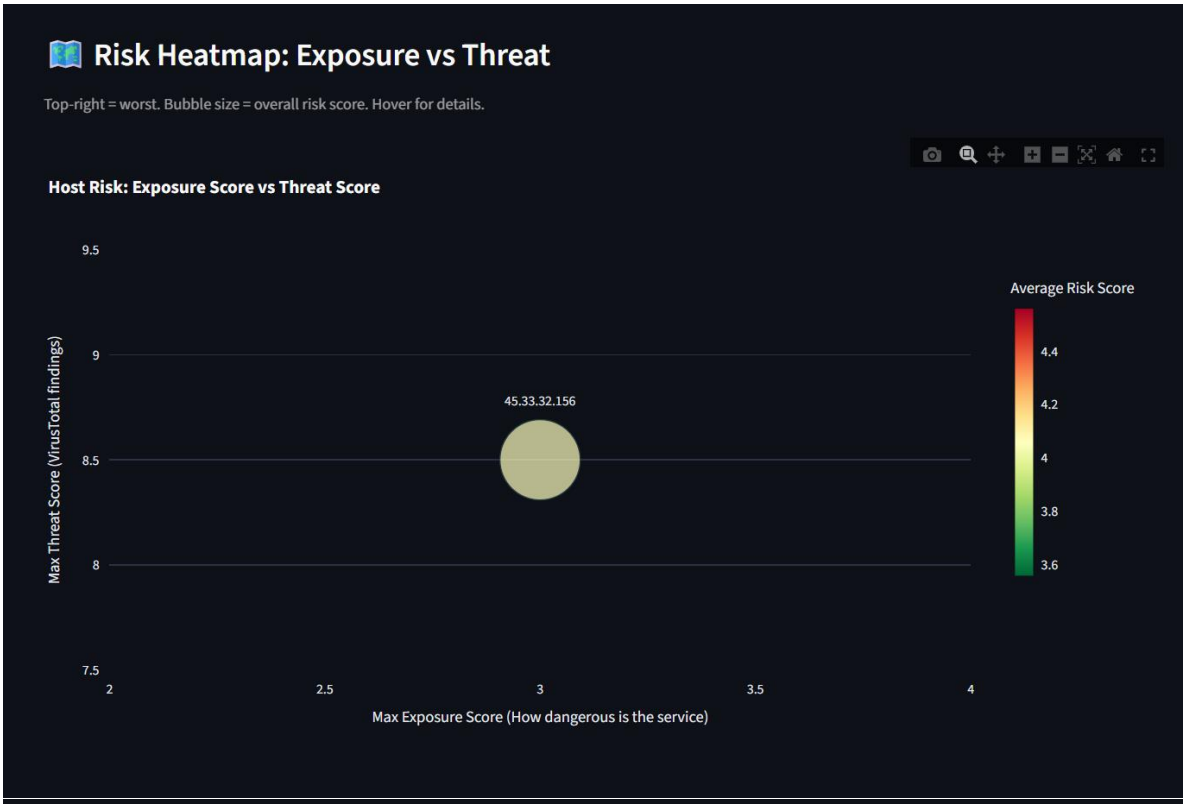
[Medium] **45.33.32.156:9929** (nping-echo) — Risk Score: **3.9**

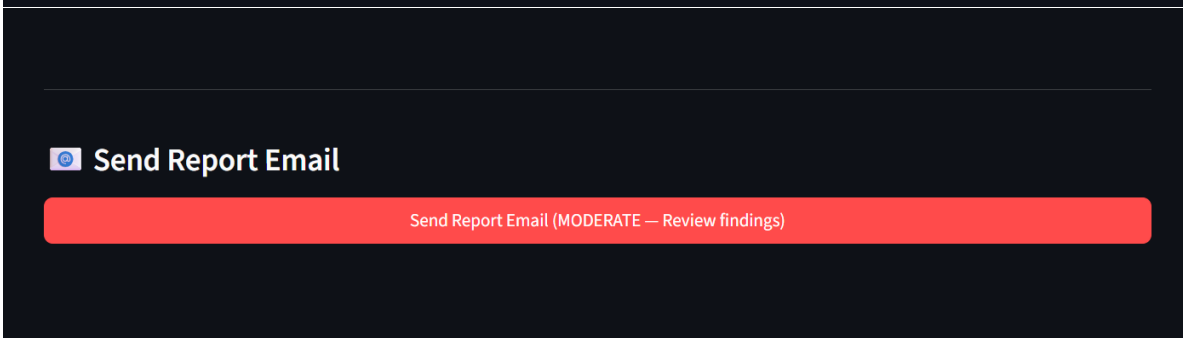
[Medium] **45.33.32.156:31337** (tcpwrapped) — Risk Score: **3.9**

Detailed Risk Overview

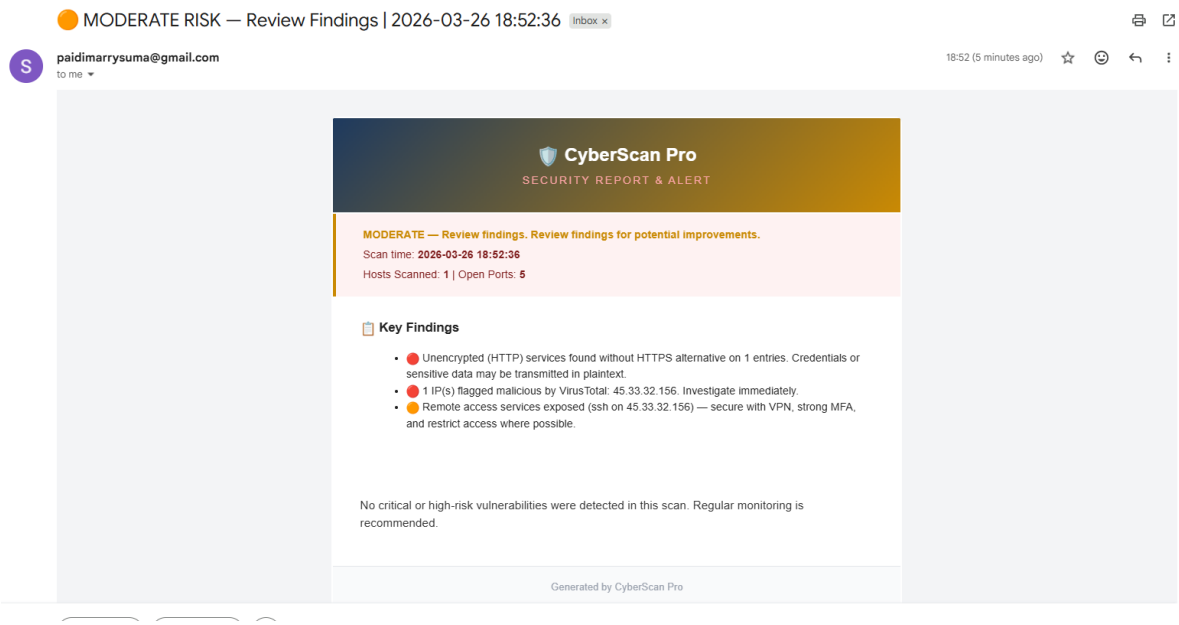
All identified ports and services, sortable and filterable.

	IP	Port	Proto	Service	Product	Version	Severity	Risk Score	VT Maliciou	VT Suspicio	Country	VT Categories
0	45.33.32.15	22	tcp	ssh	OpenSSH	6.6.1p1 Ubu	Medium	4.3	4	1	US	
1	45.33.32.15	80	tcp	http	Apache http	2.4.7	Medium	4.3	4	1	US	
2	45.33.32.15	53	tcp	domain			Medium	3.9	4	1	US	
3	45.33.32.15	9929	tcp	nping-echo	Nping echo		Medium	3.9	4	1	US	
4	45.33.32.15	31337	tcp	tcpwrappe			Medium	3.9	4	1	US	


Risk Heatmap: Exposure vs Threat
Top-right = worst. Bubble size = overall risk score. Hover for details.
Host Risk: Exposure Score vs Threat Score
Max Threat Score (VirusTotal findings)
9.5
9
8.5
8
7.5
45.33.32.156
2
2.5
3
3.5
4
Max Exposure Score (How dangerous is the service)
Average Risk Score
4.4
4.2
4
3.8
3.6


Send Report Email
Send Report Email (MODERATE — Review findings)

Email alert


MODERATE RISK — Review Findings | 2026-03-26 18:52:36
Inbox
paidimarrysuma@gmail.com
to me
18:52 (5 minutes ago)
CyberScan Pro
SECURITY REPORT & ALERT
MODERATE — Review findings. Review findings for potential improvements.
Scan time: 2026-03-26 18:52:36
Hosts Scanned: 1 | Open Ports: 5
Key Findings
Unencrypted (HTTP) services found without HTTPS alternative on 1 entries. Credentials or sensitive data may be transmitted in plaintext.
1 IP(s) flagged malicious by VirusTotal: 45.33.32.156. Investigate immediately.
Remote access services exposed (ssh on 45.33.32.156) — secure with VPN, strong MFA, and restrict access where possible.
No critical or high-risk vulnerabilities were detected in this scan. Regular monitoring is recommended.
Generated by CyberScan Pro

CONCLUSION

This project helped me understand how real-world cybersecurity tools work together to analyze risks in a system. By combining Nmap scanning with VirusTotal threat intelligence, I was able to build a complete pipeline that not only detects open ports and services but also evaluates how risky they are.

Through this project, I learned how to work with APIs, process and normalize data, and design a scoring system to represent security risks in a meaningful way. Building the interactive dashboard, email alerts, and PDF reports also gave me practical experience in developing user-focused features.

Overall, this project gave me a strong foundation in security analysis and showed me how different components—scanning, analysis, and visualization—come together to create a useful cybersecurity solution. With further improvements, this platform can be extended into a more advanced and scalable security tool.